ModelIt!

Teacher's Guide

One Species Changes Everything

How Keystone Species Hold Ecosystems Together

Quick Reference

Grade Level	Grade NE
NGSS Standard	MS-LS2-4, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS2-4, HS-LS2-6

Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how a keystone species creates a cascade effect through habitat quality, food availability, and biodiversity.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Organisms are connected through their relationships — removing a keystone species disrupts the entire web of interdependence.
Crosscutting Concept	**Stability and Change** — The ecosystem is stable when the feedback loop is active, but removing the keystone species destabilizes the entire system — demonstrating how small changes can have large effects.

CAST Testing Alignment

MULTIPLE CHOICE:

Beavers are removed from a watershed. Over the next 5 years, which change would you MOST expect to see? A) More fish in the streams B) Fewer bird species in the area C) Larger trees growing D) Cooler summer temperatures

CONSTRUCTED RESPONSE:

Using your model, explain what happens to an ecosystem when its keystone species is removed. Include the concepts of cascade effect and feedback loop in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The concept of the keystone species is one of the most important ideas in ecology. Just as a keystone holds an arch together, certain species hold entire ecosystems together — and removing them causes everything to collapse.

What Makes a Species 'Keystone'?

A keystone species has a disproportionately large effect on its ecosystem relative to its population size. Remove it, and the ecosystem changes dramatically. Not all species are keystone — some can be removed with minimal impact. The term was coined by ecologist Robert Paine in 1969 after studying starfish in tide pools.

Beavers: The Ultimate Ecosystem Engineers

Beavers are perhaps the most powerful keystone species in North America. Their dams create ponds and wetlands that support fish, amphibians, birds, insects, and plants. A single beaver family can increase local biodiversity by 30-50%. When beavers are removed, dams decay, ponds drain, and this entire web of life collapses.

Feedback Loops in Ecosystems

Ecosystems contain feedback loops — chains of cause and effect that circle back on themselves. A positive feedback loop amplifies change: more beavers → better habitat → more food → more species → even better habitat. But feedback loops work in both directions: fewer beavers → worse habitat → less food → fewer species → even worse habitat. This is why ecosystem decline can accelerate.

Trophic Cascades

When a keystone species is removed, the effects cascade through the entire food web. The classic example is wolves in Yellowstone: removing wolves → elk overpopulation → overgrazing → river erosion → loss of songbird habitat. Reintroducing wolves reversed the entire cascade. This is called a trophic cascade.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Keystone Species Population as the single external component and Habitat Quality, Food Availability, and Number of Other Species as internal components that form a cascade with a feedback loop.

E **Establish Relationships**

Students determine that Keystone Species positively affects Habitat Quality, which positively affects Food Availability, which positively affects Number of Other Species. The feedback loop: Number of Other Species positively affects Habitat Quality.

V **Visualize & Model**

Students build a model showing a linear cascade from the external input through three internal components, PLUS a feedback arrow from the last internal back to the second.

E **Evaluate Outcomes**

Students run thriving, removal, and recovery scenarios to observe the cascade and feedback loop in action.

R **Revise & Extend**

Students add Invasive Species or Human Development to see how additional pressures interact with the keystone dynamic.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Can one species really matter that much? What if I told you that removing ONE animal could turn a lush valley into a desert?"*

Watch For

Show before/after photos of landscapes with and without beaver activity. The contrast is dramatic.

Slide 2: Learning Objectives

3 min

Say: *"Today we are modeling something called a keystone species — and discovering feedback loops."*

Watch For

Pre-teach 'keystone' using the architecture analogy (show a real stone arch, identify the keystone). If the keystone is removed, the arch collapses.

Slide 3: Big Question

3 min

Say: *"Think about your favorite place in nature. What would happen if one species disappeared?"*

Watch For

Quick brainstorm: students list species they think might be keystone in their local ecosystem.

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model with a new feature today — a feedback loop."*

Watch For

Preview LEVER steps. Flag that this lesson introduces loops, which are a new modeling concept.

Slide 5: Activity 1: Components

8 min

Say: *"One external component drives THREE internal ones. Which is the input and which are the results?"*

Watch For

Guide sorting. Key discussion: why is Keystone Species the only external? Because it is the trigger for everything else.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the cascade: keystone → habitat → food → species. But here is the COOL part — species feeds BACK into habitat!"*

Watch For

Draw the cascade arrows, then add the feedback loop arrow. Discuss: What does this loop mean? It means the system builds on itself!

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us see the cascade in action. First with a thriving colony, then we pull the keystone out entirely."*

Watch For

Run all three scenarios. Key moment: the removal scenario — watch everything collapse in sequence.

Slide 8: Discoveries

5 min

Say: *"The feedback loop is what makes this so powerful — AND so dangerous. It amplifies change in BOTH directions."*

Watch For

Discuss the Yellowstone wolf reintroduction as a real-world parallel.

Slide 9: STEM Challenge

12 min

Say: *"Time to experience the cascade physically — with Jenga!"*

Watch For

Distribute labeled Jenga sets. Groups play and record which removals cause collapse.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Keystone Species Population
- **INTERNAL:** Habitat Quality, Food Availability, Number of Other Species

Relationships

- **Keystone Species Population to Habitat Quality:** POSITIVE (+) — More keystone organisms (like beavers) create and maintain habitat — dams create ponds, dig channels, flood areas that become wetlands. Their physical activity directly improves habitat quality.
- **Habitat Quality to Food Availability:** POSITIVE (+) — Better habitat supports more plant growth, more insects, more small animals — all of which are food for other species. A healthy wetland produces far more food than a dry streambed.
- **Food Availability to Number of Other Species:** POSITIVE (+) — More available food supports more species. When food is abundant, specialists and generalists alike can find niches. When food is scarce, only the toughest generalists survive.
- **Number of Other Species to Habitat Quality:** POSITIVE (+) — FEEDBACK LOOP — More species improve habitat quality — plants stabilize soil, predators control herbivore overgrazing, decomposers recycle nutrients. Biodiversity itself is a habitat-building force. This creates a REINFORCING LOOP.

Key Discoveries

- One species can support an entire ecosystem — losing it causes a cascade of decline
- Habitat quality, food availability, and biodiversity are all connected in a chain
- There is a FEEDBACK LOOP: more species improve habitat quality, which supports even more species
- Recovery takes time — ecosystems do not bounce back instantly when the keystone returns

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All species are equally important to an ecosystem"	While every species plays a role, some species have dramatically larger impacts than others. Keystone species support the structure of the entire community. Removing a common grass species might have minimal effect, but removing the beavers that created the wetland where that grass grows changes EVERYTHING.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ecosystems recover instantly when you fix the problem"	Ecosystem recovery takes years to decades, not days. Even after beavers are reintroduced, it takes time for dams to be built, ponds to form, vegetation to grow, and other species to return. The feedback loop helps, but the first cycle is the hardest to start. This is why prevention is always better than restoration.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Top predators are always the keystone species"	Keystone species can be predators (wolves), herbivores (elephants), or even small organisms (sea stars, beavers). What makes them keystone is not their size or position in the food chain — it is the MAGNITUDE of their effect on the ecosystem. Beavers are herbivores, but they reshape entire landscapes.